- TaskA: Has the highest priority. It takes the semaphore, blinks the Green light, again puts the semaphore back, and then calls vTaskDelay to go to sleep for a short time.
- TaskB: The moment TaskA goes to sleep using vTaskDelay, TaskB jumps in. It doesn't care about the semaphore; instead, it starts doing calculation that takes 100% of the CPU's space and attention.
- TaskC: Wants to take the semaphore next and blink the Blue light, but because it has the lowest priority, it isn't allowed to interrupt TaskB's calculation. It is forced to wait.
- The Timeout (Red Light): TaskC has a timer set for 200 milliseconds to get the semaphore. Because TaskB takes long time for its calculation, TaskC's timer runs out before it ever gets a turn. It skips its turn and turns on a Red warning light to show it ran out of time.